

Knowledge Engineering (SCOM7348)

Project 2

National Student Registry

Dr. Mustafa Jarrar

University of Birzeit

mjarrar@birzeit.edu

www.jarrar.info



Goals

Student who successfully finish this project will be able to:

- Represent and publish data in RDF
- Develop and share domain vocabulary in RDF Schemas
- Reuse existing vocabulary developed by others
- Develop Webpages using RDFa
- Use ORM to model RDF Schemes at the conceptual level
- Formalize RDFs (and ORM) in Description Logic, and reason about these models using DL reasoners.
- Use semantic web technology in data integration scenarios (in e-government case study)

➔ Students will be evaluated based on how much each of the above goals is achieved.

Project Description (National Student Registry)

The central management of students' profiles by the Ministry of Higher Education is becoming an urgent need in the last years. Many students in Palestine move from one university to another, and they need to transfer their academic records. Also, the ministry needs to certify the diplomas and mark sheets of students. Moreover, there is a need to centrally manage/monitor students financial aids. Therefore, the ministry has decided to build a national student registry; such that, each semester, every university will send the academic record (i.e., mark-sheet) of every student to the ministry. The ministry will then update and integrate the academic records according to the data combined from all universities into the national student registry.

The ministry wants to specify a shared ontology (in RDFs) such that all universities can exchange their students profiles in RDF format and the data can be automatically validated and integrated after that. ➔ **Build this ontology in RDFs.**

Project Details

- Each student should work alone.
- Start brainstorming on a paper to specify what a student record contains (student, university, faculty, program, course, semester, marks, averages, and the relationships between such concepts, as well as the business rules).
- Students are required to use three different mark sheets from different universities, so to ensure that the resultant RDFs model cover the relevant exchanged information needed for this national registry.
- Student are advised to first (1) develop a conceptual model using ORM for this national registry, (2) formalize the ORM schema in description logic, then (3) translate the ORM schema into RDFs.
- Further, students are required to (4) deliver two RDF files, each file representing the data in a mark sheet. Please choose mark sheets from different universities. These RDF files must commit/use the vocabulary defined the RDFs model.

Project Details

- Students are expected to re-use standard and common ontologies such as FOAF, DOAC, etc.
- Student must validate their RDFs and RDF files using the online W3C RDF validator.
- Each student must publish his/her RDFs model and the two RDF files somewhere on the Web, and only submit the links. Files must be validated before submission.
- After submitting the first version through Ritaj, Each student will be asked to present his/her models to other students next lecture, so to collect comments and feedback, and learn from each other. Then each student will have few more days to correct his/her mistakes, and submit the final model.
- **The final delivery should include:**
 - 1) the ORM schema, in PDF format
 - 2) The formalization in Description Logic, in PDF format.
 - 3) Link to the RDFs file
 - 4) Links to the two RDF files

Optional Project (**Generate DB Schema**)

Three extra marks will be added to the general average of students who: generate a database schema (out of the ORM schema) developed in Project 2.

- The database can be automatically (one click) generated by ORM tools.
- Student are advised to compare the generated DB Schema and the ORM schema, to spot mistakes, or un-intended models.
- Poor DB schema might be give low marks.
- **The final delivery should be a DB diagram , in PDF file**

Optional Project (RDFs page)

Five extra marks will be added to the general average of students who develop a webpage in RDFa:

- Students should take one of the RDF files developed in project 2 (i.e., a mark sheet) and represent this data in a webpage using RDFa technology.
- To see that what you did is correct, use the RDFa Distiller (<http://www.w3.org/2007/08/pyRdfa/>) to extract the RDF triple from your webpage. These extracted RDF triples should be same as the original RDF file.

The final delivery should be a link to the RDFa page, in PDF file.